GPU substructure for flashjet

kt / C-A jet substructure on the merge history

exclusive jets · soft-drop grooming · Lund coordinates

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Every claim below is closed against an independent reference and/or the defining paper. All plots regenerable: `plots/2026-07-13-substructure/`.

Summary — what this deck shows

Three substructure features (F1 exclusive-kt jets, F2 soft-drop/mMDT grooming, F3 Lund coordinates), all **pure-torch post-reads of the existing merge history** — no kernel changes, CPU/CUDA identical, negligible cost.

Validation ladder, each rung closed:

rung	reference	headline result
unit tests	independent NumPy tree-walks	85 passed (13 CUDA-only skipped)
paper closures	analytic predictions, toy shower	z_g on the $1/z$ curve; areas; β-ordering
real CMS QCD	stored FastJet branches (raw-to-raw)	p_T **1.000000**; m_{SD} **−0.004 GeV**; R_g 99.2% <0.01
real CMS ttbar ×2	stored FastJet branches (raw-to-raw)	dileptonic UL18 **and** fully-hadronic Run 3 close
full-event	ΔR-match to CMS's own AK8 jets	100% within 2%, median ΔR 0.0019
physics regimes	QCD vs b-jets vs boosted tops	m_W/m_t peaks, top blob in the Lund plane

flashjet reproduces CMS's FastJet reconstruction to NanoAOD storage precision. The ~4% m_{SD} tail is fully attributed (missing soft constituents + storage rounding + threshold flips — see outlier anatomy); relative agreement is ~0.1% at all masses.

Where this sits in flashjet

flashjet clusters padded `(B, N, 4)` torch tensors into jets on the GPU, staying on-device for ML training loops. Generalized-kt:

$$d_{ij} = \min(k_{t,i}^{2p}, k_{t,j}^{2p}) \frac{\Delta R_{ij}^2}{R^2}, \qquad p = -1 \text{ (anti-}k_t\text{)},\ 0 \text{ (C/A)},\ +1 \text{ (}k_t\text{)}$$

The key object Alex's kernels already produce: the *merge history*. Every recombination step records four arrays — `hist_p1, hist_p2` (the two inputs), `hist_child` (the output id), `hist_d` (the distance at which they merged). That history *is* the full binary clustering tree.

> **Our contribution = read that tree.** No new kernels, no changes to the clustering path. All three features are pure-torch post-reads of `(hist_p1, hist_p2, hist_child, hist_d)`, so they run unchanged on CPU or CUDA and add negligible cost.

What was already there vs. what we added

Pre-existing (Alex)

- `history.py`: `jet_idx_from_history`, `_resolve_parents`, `_decode`, `splitting_scales_from_history`, Triton `_decode_triton`
- `reference.py`: single-event NumPy ground truth (`cluster_event`)
- `nn_reference.py`: nearest-neighbour reference
- `conftest.py`: `random_event` fixtures
- The kernels + the merge-history recording

Added (this work) — in `history.py`

- **F1** `exclusive_jets_from_history`
- **F2** `groom_from_history`
- **F3** `lund_coordinates_from_history`
- helpers: `_resolve_roots`, `_jet_roots`, `_pseudojet_p4`, `_dense_number_roots`
- `api.py`: `ClusterOutput.{exclusive_jets, groomed_jets, mass_drop, lund_coordinates}` + a `mask` field
- `tests/test_substructure.py` (new)

The `_ref_*` walkers inside `test_substructure.py` are also ours — independent naive NumPy tree-walks used only to pin the fast implementations.

The three features

	Feature	Function	Implements
F1	Exclusive jets (kt)	`exclusive_jets_from_history(...)`	kt algorithm — FastJet manual [1111.6097], [0802.1189]
F2	Grooming (C/A)	`groom_from_history(...)`	Soft Drop [1402.2657] · mMDT [1307.0007] · mass-drop [0802.2470]
F3	Lund coordinates	`lund_coordinates_from_history(...)`	Primary Lund plane [1807.04758]

F1 — undo the last merges of the sequence to expose exactly `n_jets` (or a `d_cut`) subjets; reduces to the inclusive jets at the trivial cut.

F2 — walk each jet down the harder branch, dropping the softer prong until $z > z_{\text{cut}}(\Delta R/R)^{\beta}$ (soft-drop) — the $O(\log_2 n)$ declustering.

F3 — for every primary split emit $(z, \Delta R, k_t, \ln 1/\Delta R, \ln k_t, d)$; the d-channel ties **exactly** to `splitting_scales`.

How the toys were generated

(the "simulation" — none of it pre-existing; flashjet only clusters, it does not generate events)

The toy event generators (written in the plot scripts)

flashjet takes 4-vectors in and returns jets — it has **no event generator**. To exercise the features I wrote small, transparent toys *outside the repo*, in the plot scripts.

`make_plots.py` — two single-fat-jet samples at $p_T \approx 500$ GeV, $R = 0.8$:

- **QCD-like**: one hard collinear core (92% of p_T, $\sigma_{y,\phi} = 0.06$) + soft wide-angle radiation (8%, $\sigma = 0.30$). Each "spray" = n massive pions with exponential p_T fractions.
- **W-like**: two hard prongs with pair mass $m = \sqrt{z(1-z)}\, p_T\, \Delta R = 80.4$ GeV ($z \in [0.30, 0.45]$), random orientation, + 6% soft contamination.

```
def _spray(rng, n, y0, phi0, pt_total, sigma):     # n pions around (y0,phi0)
    frac = rng.exponential(1., n); frac /= frac.sum()
    pt = pt_total * frac
    y, phi = rng.normal(y0, sigma, n), rng.normal(phi0, sigma, n)
    mt = np.sqrt(M_PI**2 + pt**2)
    return np.stack([pt*np.cos(phi), pt*np.sin(phi), mt*np.sinh(y), mt*np.cosh(y)], 1)
```

1500 events per sample. Purpose: QCD vs W is a *known* truth, so any observable that fails to separate them (or lands off the analytic curve) would be a real bug.

The toy parton shower (for the paper-figure closures)

`make_paper_plots.py` builds a **primary, fixed-coupling, leading-log** shower — emissions sampled **uniformly in the Lund triangle** with density $\bar{\alpha}$ per unit $(\ln 1/\theta,\ \ln k_t)$ area, off a hard spine at $(y, \phi) = (0, \pi)$:

$$\theta = e^{-u}, \quad k_t = e^{v}, \quad z = \frac{k_t}{p_{T0}\,\theta}, \qquad \text{keep } v \le \ln(p_{T0}/2) - u\ (z \le \tfrac{1}{2}),\ k_t > k_t^{\min}$$

This is **exactly the semi-classical picture the papers' analytic predictions are derived in** — which is what makes the closure *quantitative*, not just qualitative.

- **Jet areas**: one hard event + a dense grid of infinitely-soft **ghosts** ($p_T = 10^{-8}$); the ghosts each algorithm sweeps up trace its catchment area (the anti-kt-paper construction).

$\bar{\alpha} = 0.25$, $p_{T0} = 1$ TeV, $R = 0.4$, 20 000 showers. Everything is re-runnable with a fixed seed (`20260713`).

Correctness

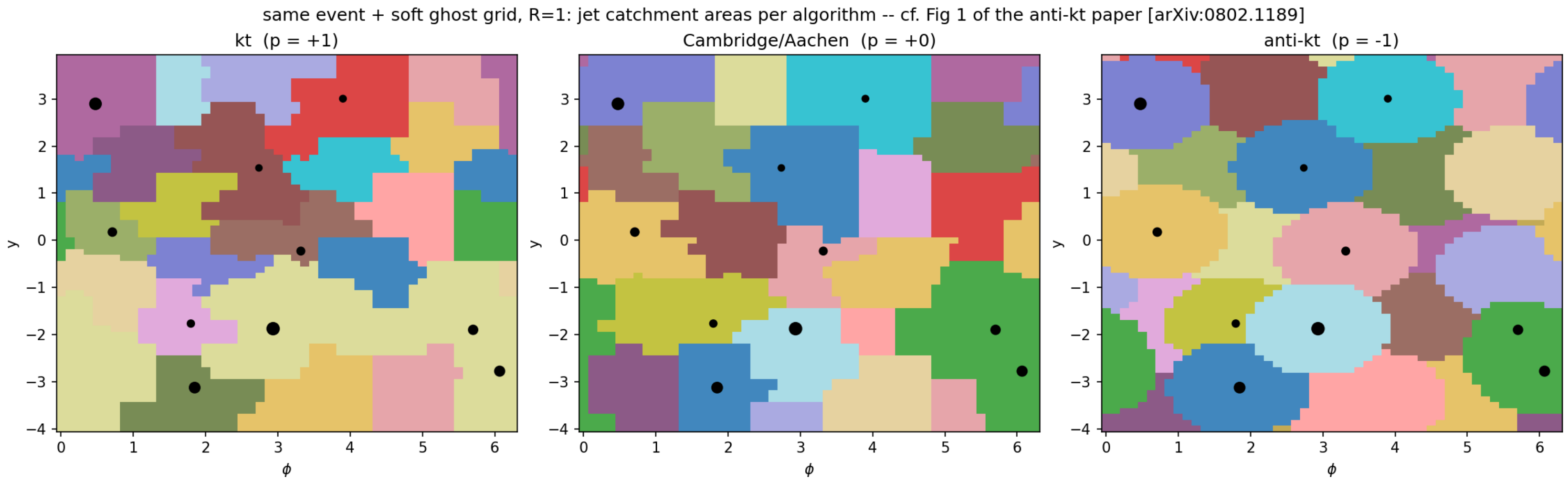
each feature reproduces the signature figure of the paper it implements

Every slide states its **input**. Three input classes appear:

(A) ad-hoc QCD/W toys (`make_plots.py`), **(B)** a leading-log toy shower (`make_paper_plots.py`),

(C) real CMS PF-candidate constituents (`make_cms_plots.py`). All seed `20260713`.

anti-kt jet shapes — reproduces [0802.1189] Fig. 1

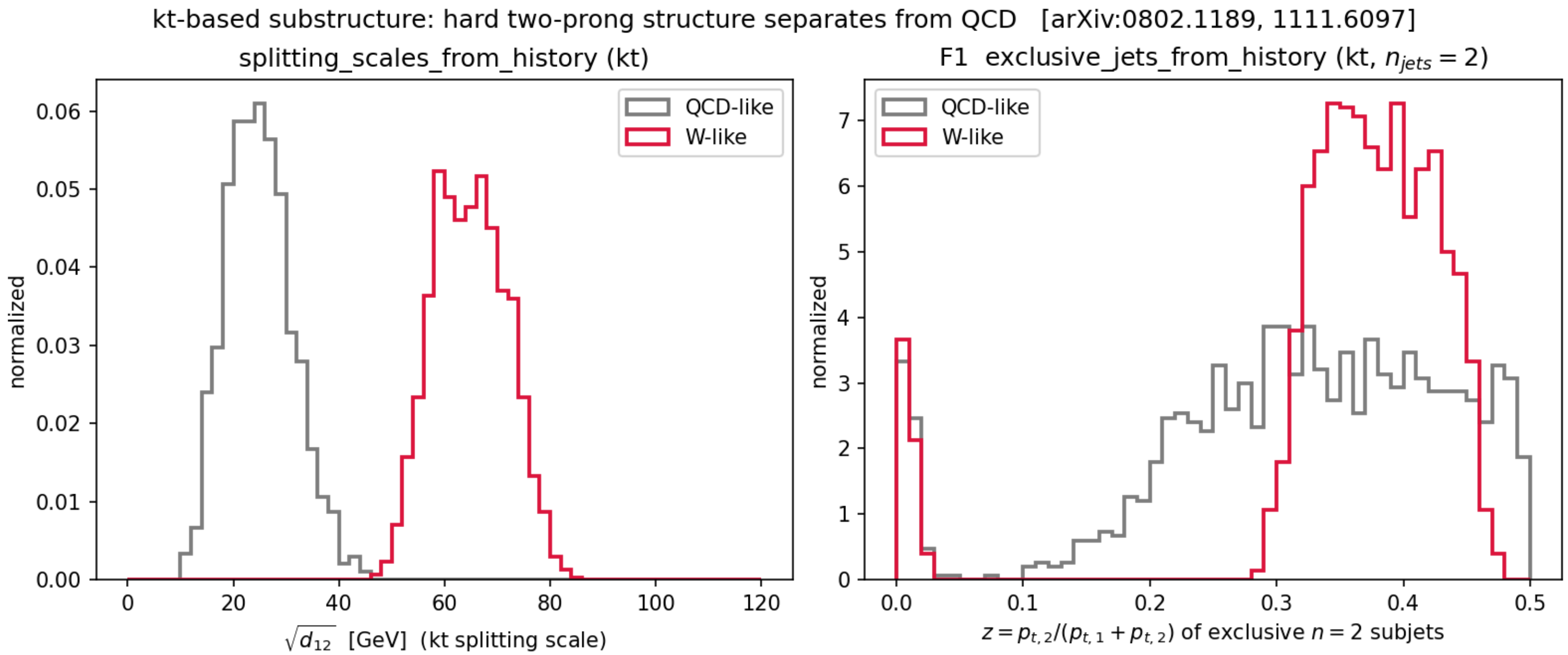


What: each algorithm's **jet catchment area**. kt & C/A give ragged, area-fluctuating jets; **anti-kt** gives rigid **circles** around each hard particle — the defining anti-kt result, from flashjet's own clustering.

How made — input (B), no substructure feature, clustering only.

1 synthetic event: **10 hard massless particles** (p_T 10–400 GeV, random y, ϕ) + a **uniform grid of ~3200 "ghosts"** ($p_T = 10^{-8}$, spacing 0.125 in y, ϕ). Clustered at $R = 1$ with `cluster_event_nn` for $p = +1/0/-1$; each ghost is coloured by the jet it lands in.

F1 — kt substructure separates 2-prong from QCD [0802.1189, 1111.6097]



Left — $\sqrt{d_{12}}$ (`splitting_scales_from_history`): kt scale of the last merge.
W-like peaks at $\sim m_W/2$, QCD low.
Right — exclusive 2-subjet z
(`exclusive_jets_from_history`, $n_{\text{jets}} = 2$):
W-like balanced ($z \approx 0.35$), QCD lopsided ($z \to 0$).

How made — input (A), leading jet, kt clustering.
1500 **QCD-like** + 1500 **W-like** toy fat-jets ($p_T \approx 500$ GeV, $R = 0.8$); each a bag of
massive-pion 4-vectors from `_spray` — **synthetic particles, not real constituents**. W pair-mass
= 80.4 GeV. Both observables read the **kt** merge history.

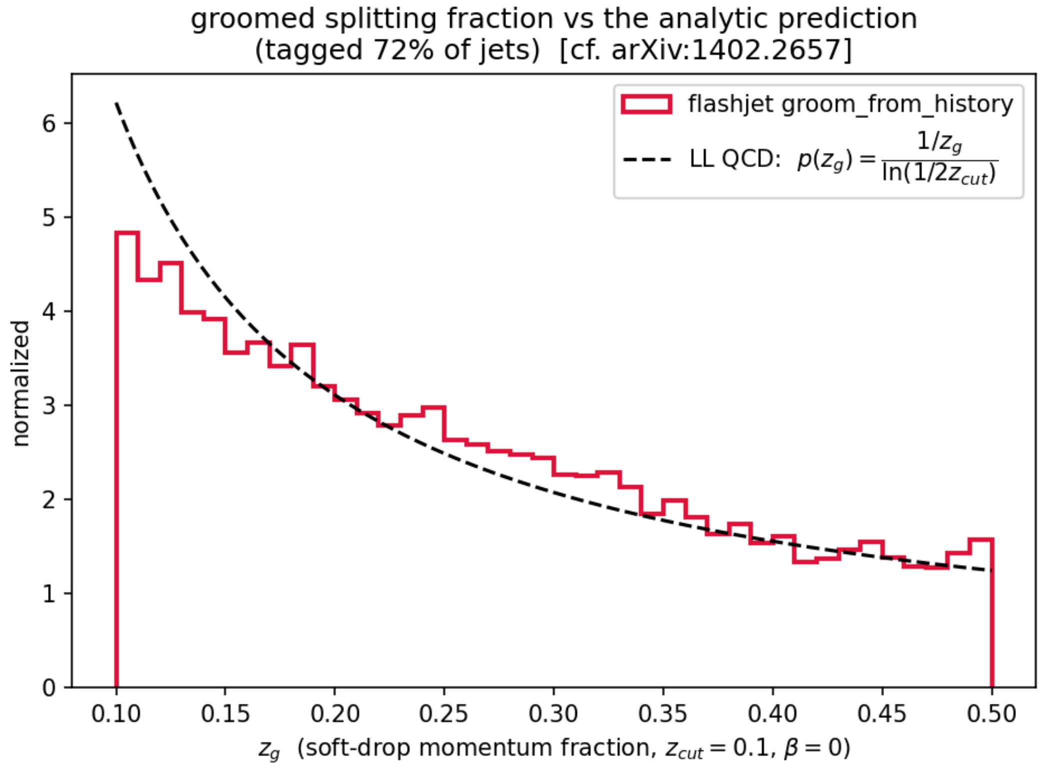
F2 — soft-drop z_g vs the analytic prediction [1402.2657]

The soft-drop momentum fraction z_g from `groom_from_history` ($z_{\text{cut}} = 0.1,\ \beta = 0$) tracks the **leading-log QCD prediction**

$$p(z_g) = \frac{1/z_g}{\ln(1/2z_{\text{cut}})}$$

across the **entire range**, no free parameters. 72% of jets tagged.

> **Decisive grooming-correctness plot**: the target curve is *the* right
> answer, so lying on it is unambiguous closure.



How made — input (B). 20 000 toy-shower jets ($p_{T0} = 1$ TeV, $R = 0.4$), leading jet
per event, soft-dropped by F2; the black curve is the analytic $1/z$ with *no* fit.

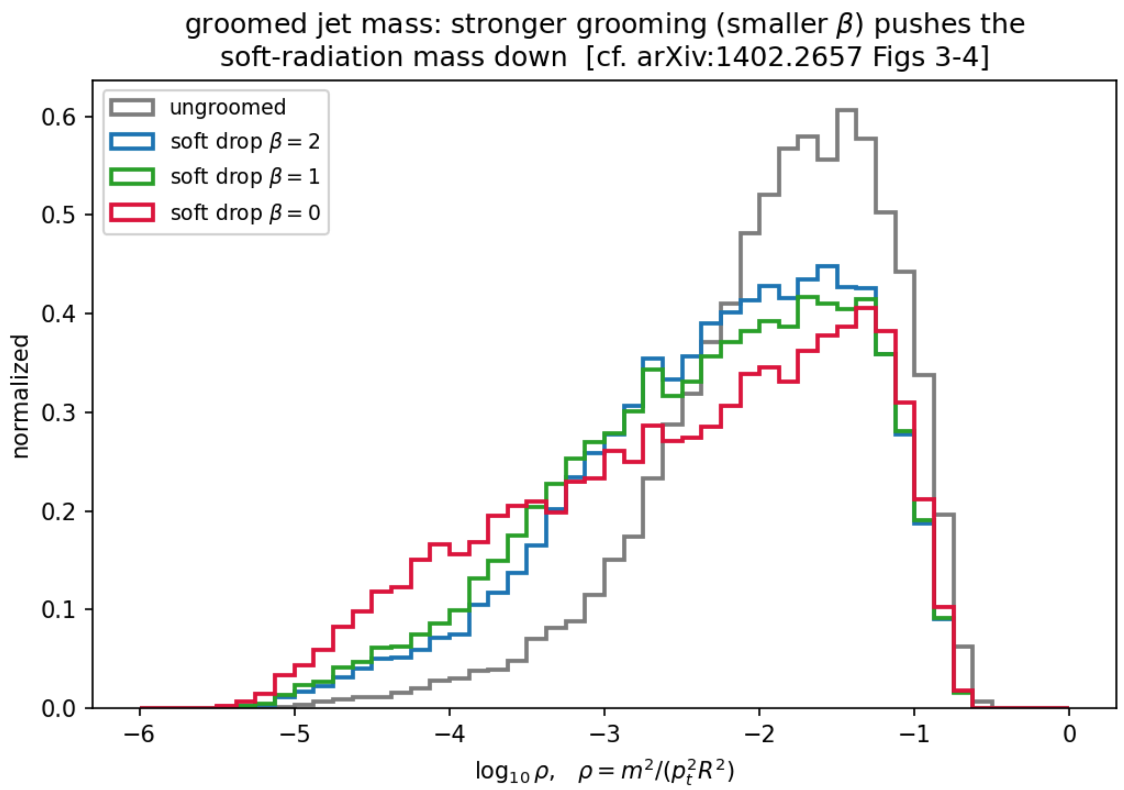
F2 — groomed mass ordering in β [1402.2657 Figs. 3–4]

Groomed mass $\rho = m^2/(p_T^2 R^2)$ for $\beta = 0, 1, 2$ vs ungroomed.

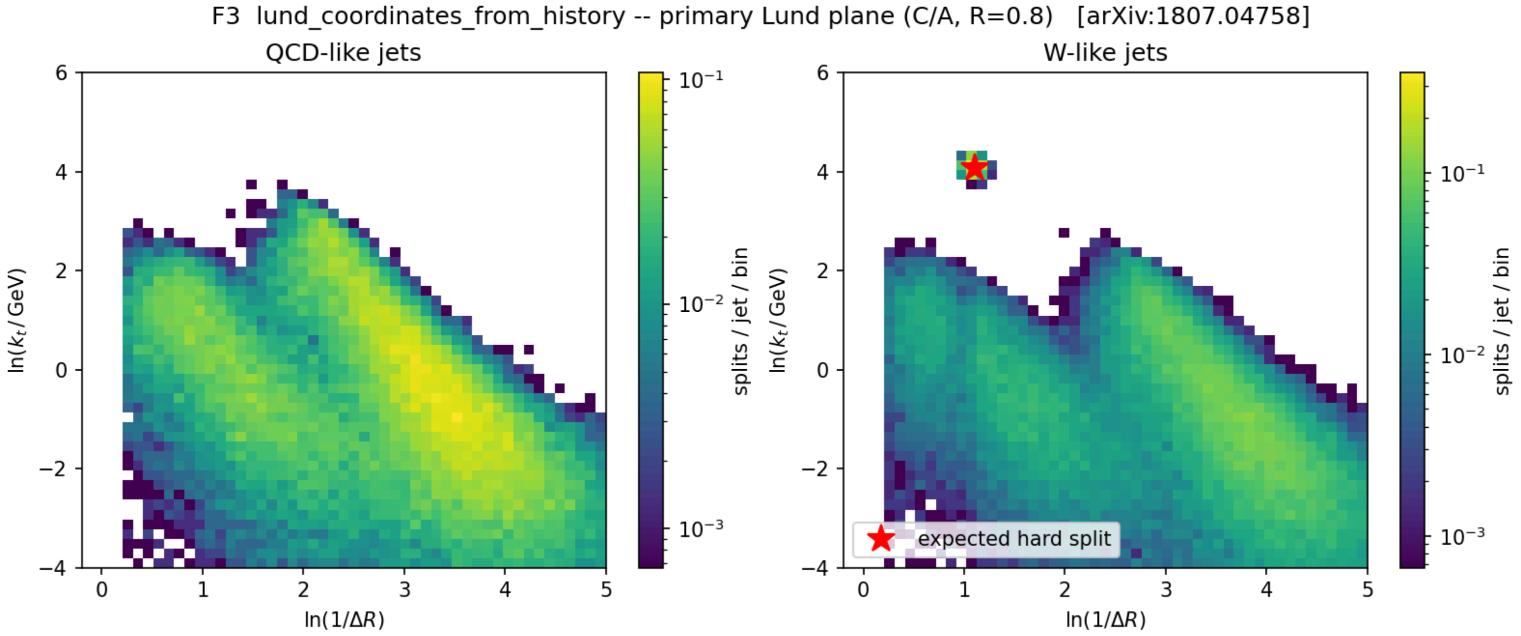
- Grooming shifts the spectrum to **lower** ρ (removes soft-wide radiation).
- **Smaller β grooms harder**: the $\beta = 0$ curve reaches furthest left.

Reproduces the β-ordering of the Soft Drop paper.

How made — input (B). Same 20 000 toy-shower jets; F2 `groom_from_history` re-run at $\beta = 0, 1, 2$ (all $z_{\text{cut}} = 0.1$) on the leading jet, ρ from the groomed 4-vector.



F3 — primary Lund plane [1807.04758]



`lund_coordinates_from_history` (C/A, $R = 0.8$). **QCD** fills the soft-collinear region smoothly. **W-like** shows the same background **plus an isolated hard-splitting spot** exactly where the 2-body m_W decay must sit (red ★ = predicted position).

How made — input (A). Same 1500+1500 QCD/W toy fat-jets as the F1 slide, **C/A** clustered;
F3 emits $(\ln 1/\Delta R,\ \ln k_t)$ for every primary split of the leading jet, histogrammed over all jets.

On real CMS data — input (C)

not toys any more: real detector-level PF-candidate constituents

The real-data pipeline (`make_cms_plots.py`)

Sample: `QCD_Pt-15to7000_Flat2018_pythia8`, **UL18 JMENano** (150X reprocessing) — the one NanoAOD flavour that stores the `PFCand` table + the `FatJetPFCand` constituent→AK8 map. Pulled via DAS/`xrdcp` → `data/qcd_jmenano_150x.root` (1.7 GB). Ordinary NanoAOD has no constituents and is useless here.

These plots use *real jet constituents*, not toys:

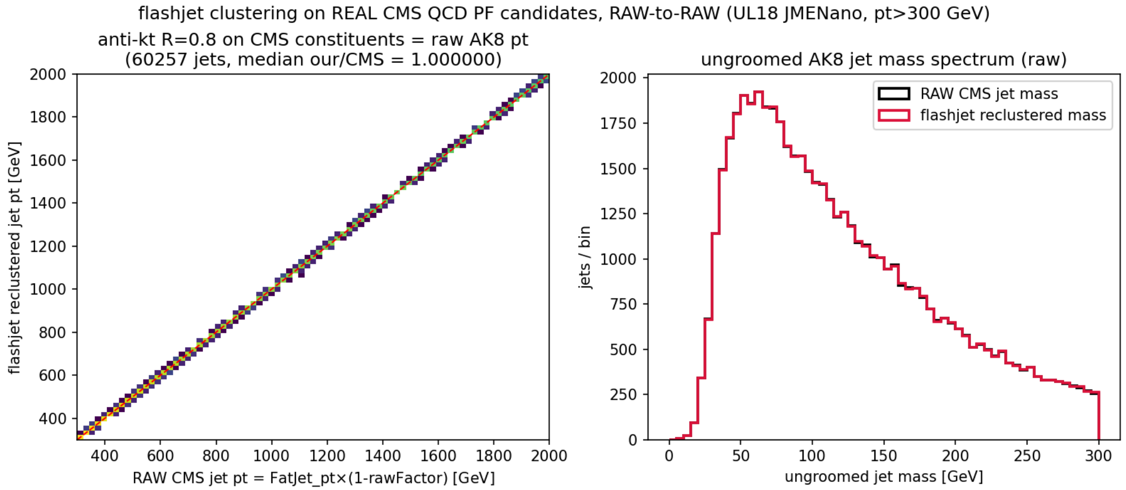
1. Pick every CMS **AK8 FatJet** with $p_T > 300$, $|\eta| < 2.4$.
2. Gather its **PF candidates** via `FatJetPFCand_jetIdx→pfCandIdx→PFCand_{pt,η,φ,m}`. **`PFCand_pt/mass` are already PUPPI-weighted** (the branch titles say so) — the weighted 4-vectors are stored directly, which is why no separate weight branch exists.
3. Feed *those constituents* to **our** `cluster(R=0.8, antikt)` — CMS AK8 **is** anti-kt $R = 0.8$.
4. Leading jet → **F2** soft drop ($z_{\text{cut}} = 0.1, \beta = 0$, declustering the **C/A** tree, as FastJet's SoftDrop does) — and **F3** Lund.
5. Chunked at 3000 jets (torch backend is $O(N^3)$; each AK8 jet is one independent event, so chunking is exact).

The stored CMS values are JEC-corrected: raw jet = `FatJet_pt×(1−rawFactor)`; `FatJet_msoftdrop` = m(sub1+sub2) **with subjet JECs** (proven from data: Δ = +0.0002 GeV).

CMS (1) — reclustering closes: our p_T vs CMS `FatJet_pt`

What: feed CMS's own AK8 constituents to **our** anti-kt $R = 0.8$ and compare the reclustered jet p_T to CMS's stored `FatJet_pt`, **jet-by-jet**.

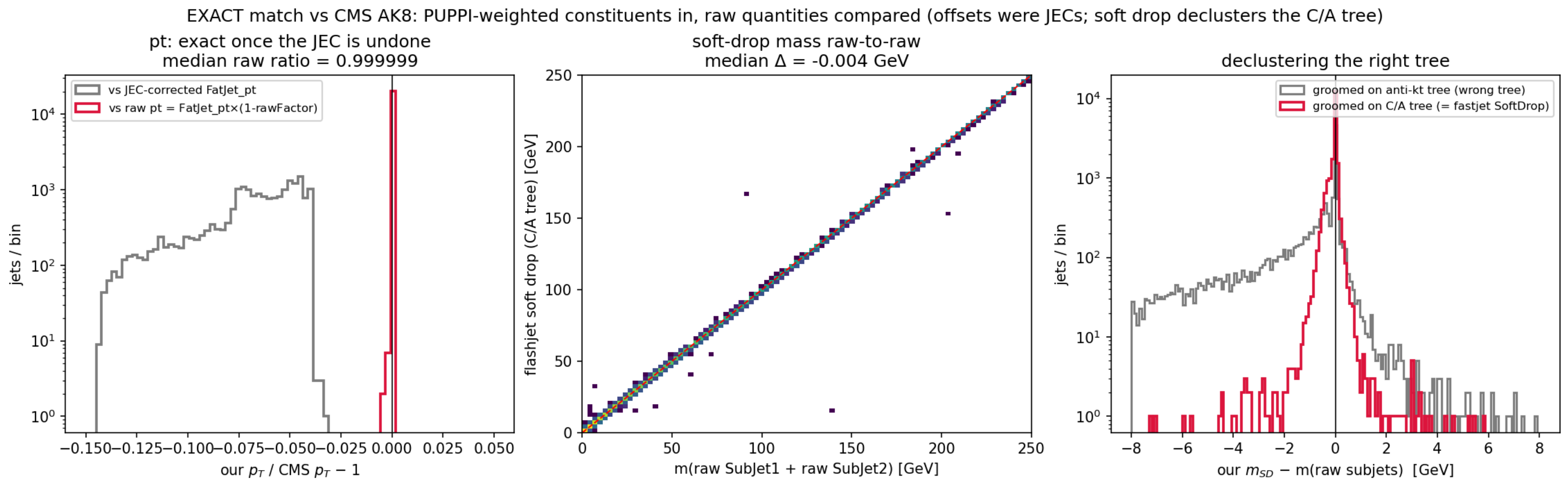
Result — EXACT: vs the **raw** jet pt (`FatJet_pt×(1−rawFactor)`) the ratio is **median 1.000000, σ = 2.5×10^{-4}** — pure NanoAOD storage precision. The apparent ~6% offset vs the stored value **is the L1L2L3 JEC**, nothing else: CMS stores corrected p_T, we recompute the raw one.



Input (C). 2D raw-p_T histogram + raw-mass spectra, real constituents (60 257 jets);

run on HTCondor (cluster 9087059).

CMS (2) — soft-drop mass matches CMS EXACTLY (raw-to-raw)



The "offset" was never PUPPI — the constituents are *already* PUPPI-weighted. Two stacked artefacts, both removed: **(1) JEC** — `FatJet_pt` / `msoftdrop` are JEC-corrected; raw jet = `pt×(1−rawFactor)`, `msoftdrop` = m(JEC-corrected sub$_1$+sub$_2$), proven from data (Δ=+0.0002 GeV).

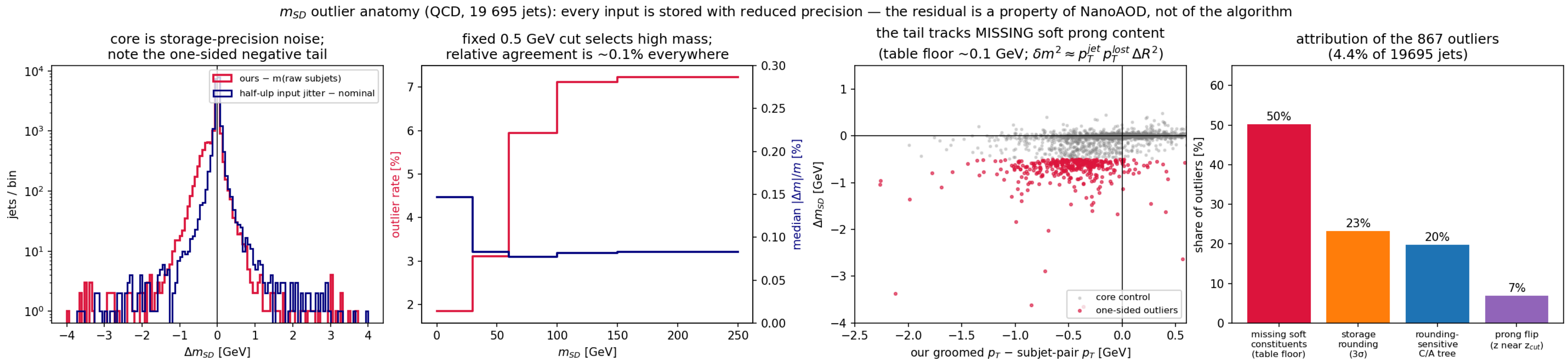
(2) Wrong tree — we groomed the anti-kt history; FastJet SoftDrop declusters a **C/A** reclustering.

Raw-to-raw, C/A tree (20 065 jets):

	our − CMS
p_T ratio	**0.999999** (σ 2×10^{-4})
m_{SD}	**−0.004 GeV** (95.6% <0.5 GeV)
z_g	|Δ| = **7×10^{-5}**

Residual = NanoAOD float precision. **F2 reproduces CMS `msoftdrop`.**

CMS (2b) — anatomy of the residual m_{SD} tail (the 4.4%)



Every input branch is stored with reduced mantissa (`PFCand_pt` ~10 bits ≈ 10^{-3}, `SubJet_pt/mass` ~9 bits, `SubJet_rawFactor` ~5 bits ≈ 2%) while CMS ran FastJet at full precision.

Attribution (per-jet tests, 19 695 jets): **50%** — soft candidates **missing from `FatJetPFCand`** (the table has an effective ~0.1 GeV floor; the tail is one-sided, its groomed-p_T deficit correlates with Δm, and $\delta m^2 \approx p_T^{jet} p_T^{lost} \Delta R^2$ matches); **23%** within 3σ of storage rounding; **20%** rounding-sensitive C/A trees (half-ulp jitter moves them); **7%** genuine $z \approx z_{\mathrm{cut}}$ prong flips (14× enriched vs core). **Not fixable from NanoAOD — and not an algorithm error**: the fixed 0.5 GeV cut just selects high-mass jets; relative agreement is ~0.1% everywhere.

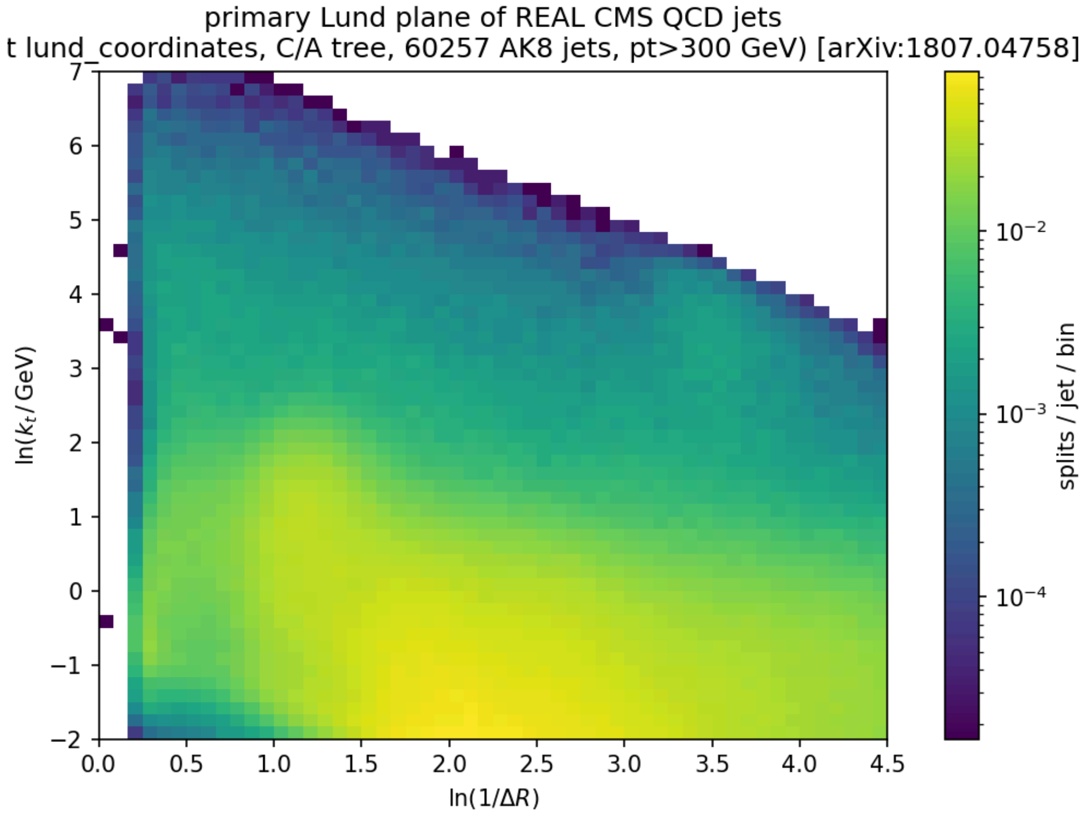
Scripts: `outliers.py`, `analyze_outliers*.py`, `rerun_rest.py` (HTCondor 9098953).

CMS (3) — primary Lund plane of 60 257 real QCD jets

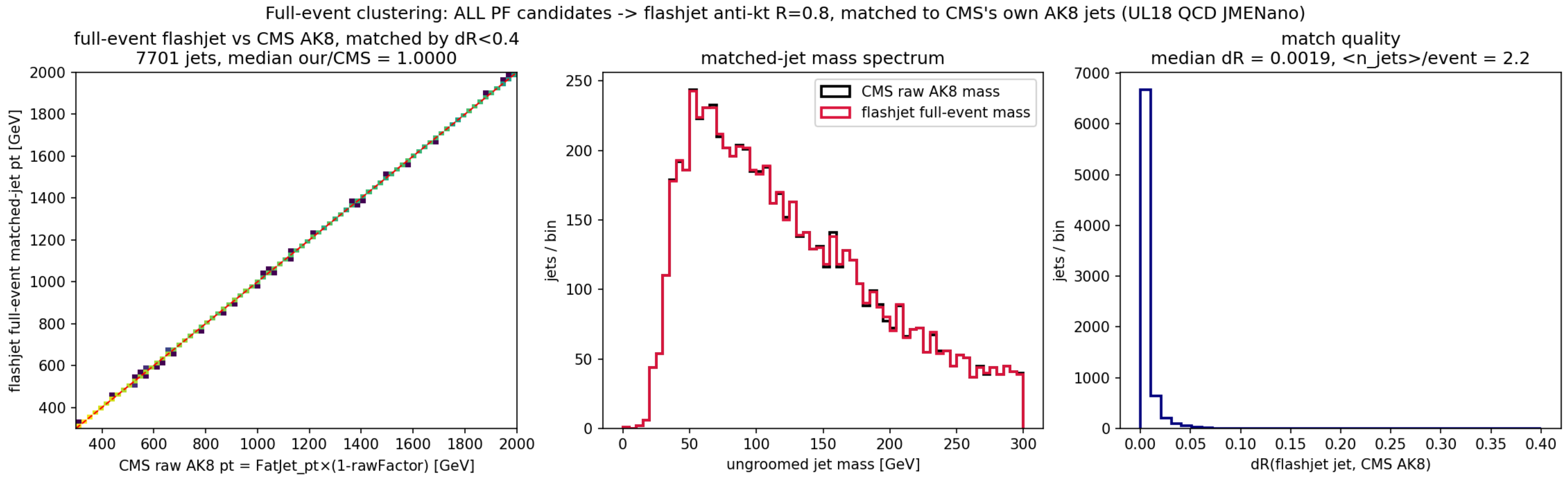
What: F3 `lund_coordinates` on the same real AK8 jets. Needs no comparison curve — it *is* a clean, publication-quality primary Lund plane straight from detector-level simulation.

The full [1807.04758] structure emerges with **no toy input**: the hard-collinear perturbative ridge, the soft plateau, and the three kinematic edges.

Input (C). 60 257 AK8 jets, $p_T > 300$ GeV, **C/A** reclustering; every primary split of each jet histogrammed.



CMS (4) — full-event clustering recovers CMS's own AK8 jets



What: the realistic path — feed **every PF candidate in the event** to flashjet anti-kt $R = 0.8$ (no `FatJetPFCand` per-jet pre-grouping), then match the resulting inclusive jets to CMS's **stored** AK8 jets (`FatJet_*`, which CMS clustered with FastJet) by $\Delta R < 0.4$.

Result (7 701 matched jets, $p_T > 300$, raw-to-raw): pt **median 1.0000, 100% within 2%**; match ΔR **median 0.0019**; mass spectra identical. Full-event flashjet reproduces CMS's own FastJet AK8 reconstruction to milliradian ΔR — no jets known a priori.

On a ttbar sample

a second, independent final state · comparison against the sample's own stored FastJet branches

CMS `TTTo2L2Nu` UL18 JMENano 150X, 344 k events → 12 561 leading AK8 jets ($p_T > 300$, $|\eta| < 2.4$).

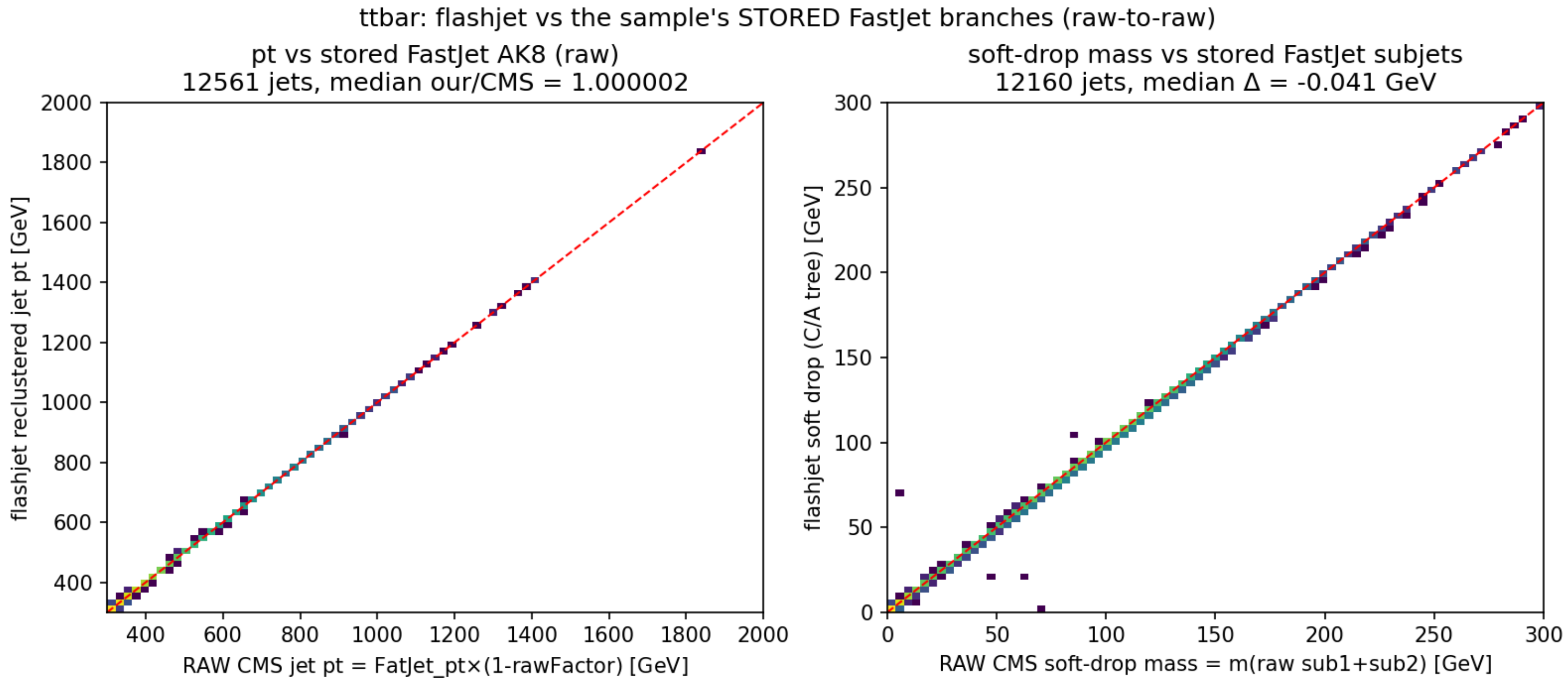
"Comparison against FastJet" = comparison against CMS's **stored** `FatJet_*` / `SubJet_*` / `tau*` branches

(CMS clustered them with FastJet), **not** a FastJet re-run.

Note: `TTTo2L2Nu` is **dileptonic** (both $W \to \ell\nu$), so these AK8 jets are mostly **b-jets + ISR**,

not hadronic top decays — a fully-hadronic sample follows in the next section.

ttbar — flashjet = stored FastJet branches (raw-to-raw)



pt (our anti-kt $R = 0.8$ vs `FatJet_pt×(1−rawFactor)`): median **1.000002**, σ 2.2×10^{-4}.

soft-drop mass (our C/A tree vs m(raw sub_1+sub_2)): median Δ **−0.041 GeV**, 94.2% <0.5 GeV.

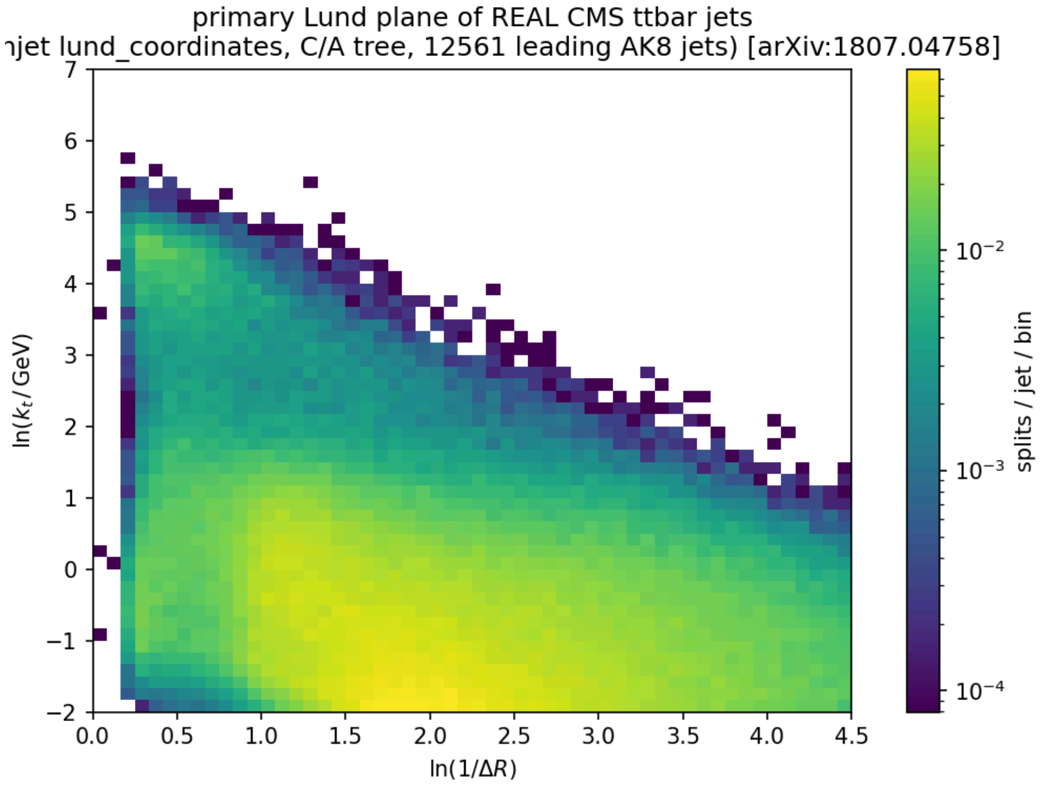
The exact match holds on a completely different final state (boosted tops + b-jets), not just QCD — same conclusion, same NanoAOD-precision residual.

ttbar — primary Lund plane (F3)

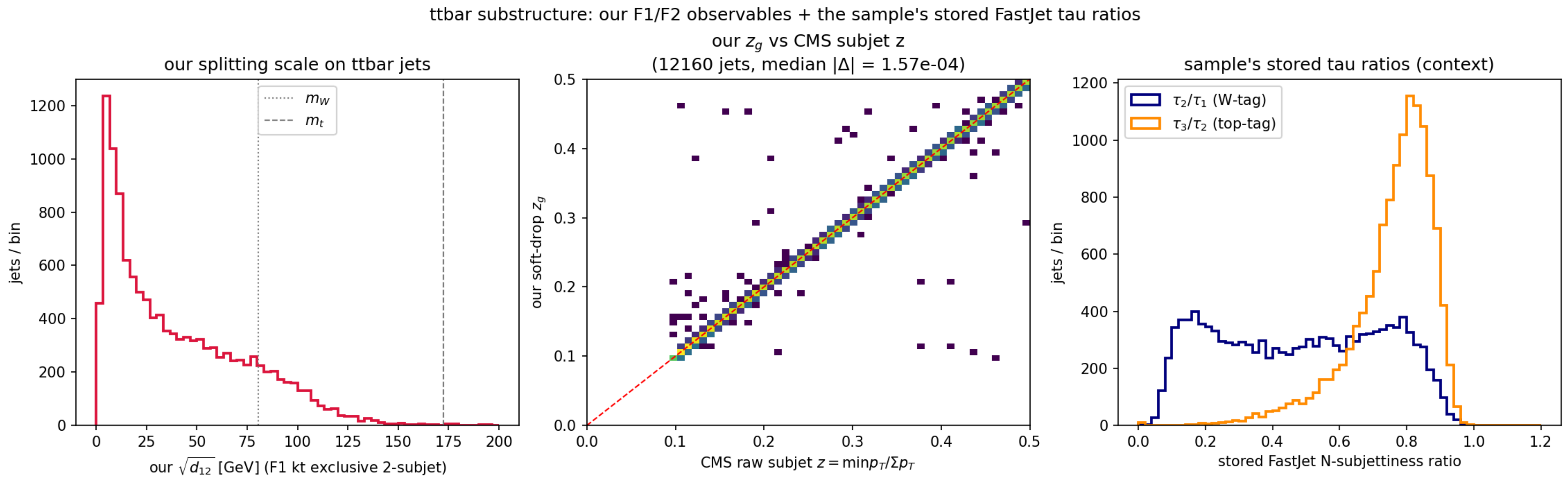
`lund_coordinates_from_history` (C/A, $R = 0.8$) on 12 561 real ttbar AK8 jets.

The full [1807.04758] structure appears, and — unlike QCD — an **enhanced hard-splitting region** around $\ln 1/\Delta R \approx 1,\ \ln k_t \approx 1$. Since this sample is **dileptonic** (no hadronic top/W decays), the enhancement comes from the different jet composition — **b-jets from the top decays + ISR** — not from prongy top decays (those appear in the fully-hadronic sample, next section).

Clean, publication-quality Lund plane straight from detector-level ttbar simulation, entirely from our F3.



ttbar — our substructure vs stored FastJet observables



Left (F1): our exclusive-2-subjet $\sqrt{d_{12}}$ (kt splitting scale) on ttbar jets; the m_W/m_t lines are **mass-scale references only** — a decay splitting sits at $\sqrt{d_{12}} = m\sqrt{z/(1-z)} \approx m/2$, and this dileptonic sample has no hadronic decays.

Middle (F2): our soft-drop z_g vs CMS's **raw subjet** z, jet-by-jet — median |Δ| = **1.6×10⁻⁴**.

Right (context): the sample's stored FastJet **N-subjettiness** ratios — τ_2/τ_1 (W-tag), τ_3/τ_2 (top-tag). Our z_g tracking CMS's stored subjet z to 10^{-4} closes F2 directly against FastJet's SoftDrop output.

Three samples side by side

*QCD vs dileptonic ttbar vs **fully-hadronic** ttbar — the first sample with real boosted W/top decays*

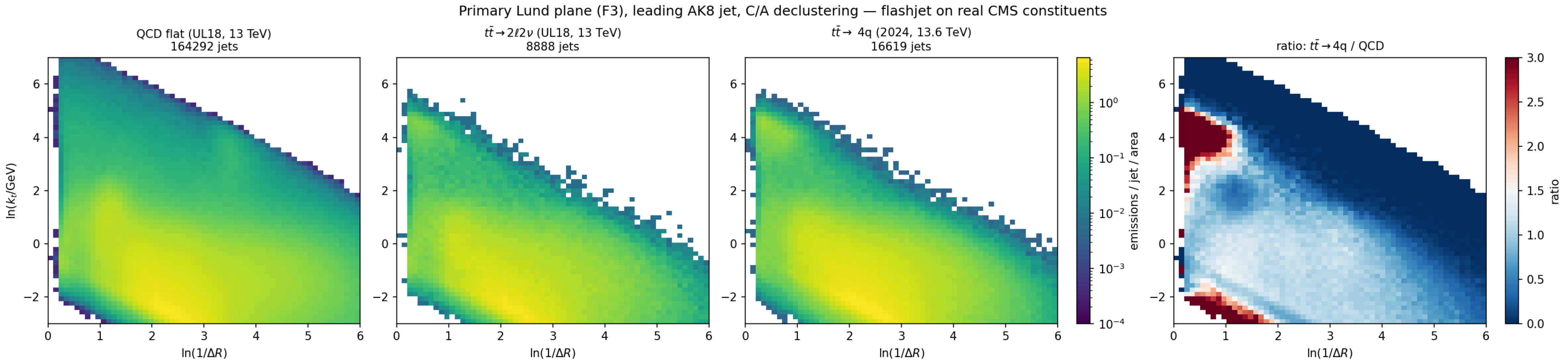
(1) `QCD_Pt-15to7000_Flat2018` UL18 JMENano (13 TeV) — 160 393 jets ·

(2) `TTTo2L2Nu` UL18 JMENano (13 TeV) — 8 639 jets ·

(3) `TTto4Q` **RunIII2024Summer24 JMENanoV15** (13.6 TeV, 2024 pileup) — 16 516 jets.

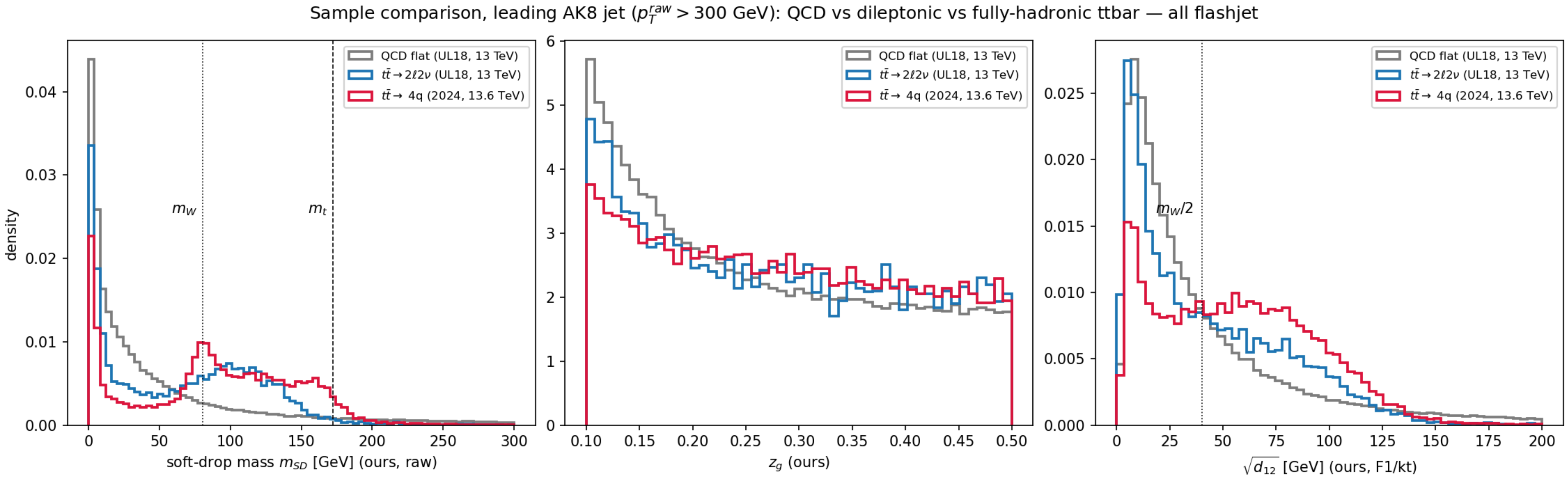
All: leading AK8 jet/event, raw $p_T > 300$ GeV, $|\eta| < 2.4$, ≤200 constituents, full files (`make_compare_plots.py`, HTCondor 9099026). Same schema everywhere: PUPPI-weighted `PFCand` + `FatJetPFCand` map.

Lund planes: QCD → b-jets → boosted tops (F3)



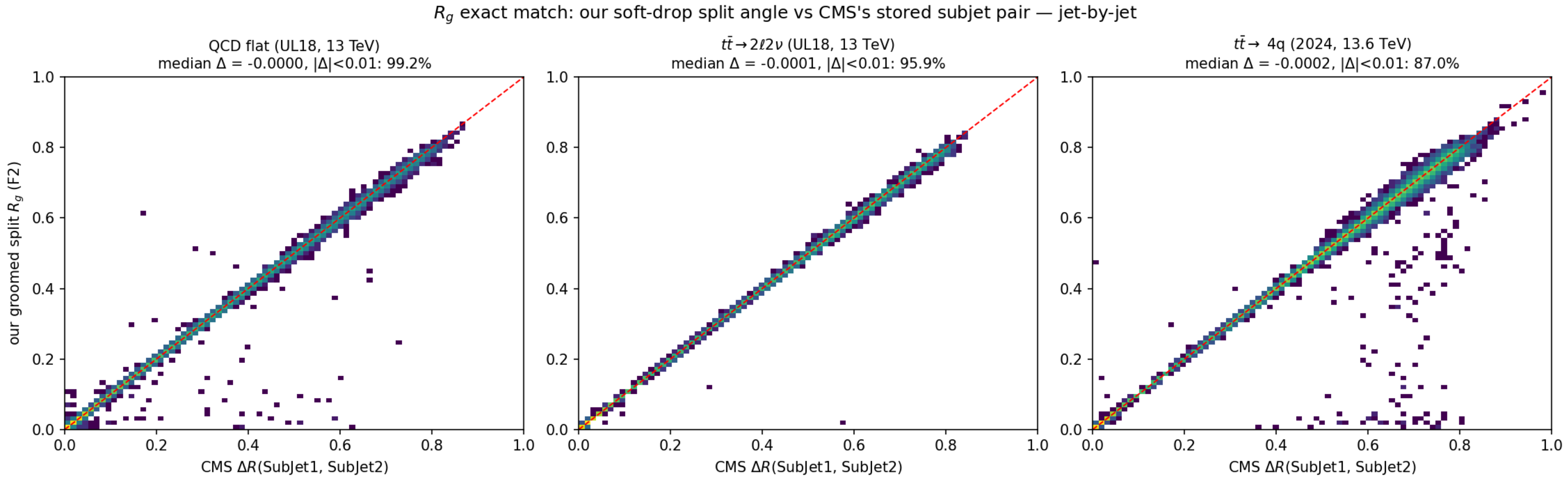
The top-decay scale appears exactly where it must: the $t\bar{t} \to 4q$ plane grows a hard-splitting blob at $\ln k_t \approx \ln(m_W/2) \approx 3.7$, wide-angle ($\ln 1/\Delta R \lesssim 1$) — over **2×** QCD in the ratio panel (right). The dileptonic sample (b-jets, no hadronic top) shows only a mild version. Same clustering, same F3, three physics regimes.

Spectra: m_W / m_t peaks from our grooming (F1 + F2)



Left (F2): our raw soft-drop mass — the fully-hadronic sample peaks at m_W with a top shoulder at m_t; dileptonic peaks broad and lower (b + extra radiation); QCD falls. **Middle (F2):** z_g — ttbar flatter than QCD's $\sim 1/z$ (hard 2-body splits). **Right (F1):** kt splitting scale $\sqrt{d_{12}} = m\sqrt{z/(1-z)}$ — checked in m_{SD} slices: W-window jets peak at **39** ≈ $m_W/2$, top-window jets at **81** ≈ $m_t/2$ (which coincidentally $\approx m_W$!); QCD collapses to low scales.

R_g — a second jet-by-jet exact match (F2 vs stored subjets)



What: our split angle R_g (`groom_from_history`'s `dR`) vs stored ΔR(sub$_1$,sub$_2$), **jet-by-jet**: median Δ ≤ 2×10^{-4} — with z_g, both grooming observables close.

Run 3 = the outlier-anatomy mechanism at a higher dose: same floor/precision, but 2024 pileup →

90% of jets have p_T ratio < 1 (UL18: 50%), **99%** of Δm outliers negative; relative agreement

stays **0.14%** — its jets are just heavy (median 93 GeV), so the 0.5 GeV window bites.

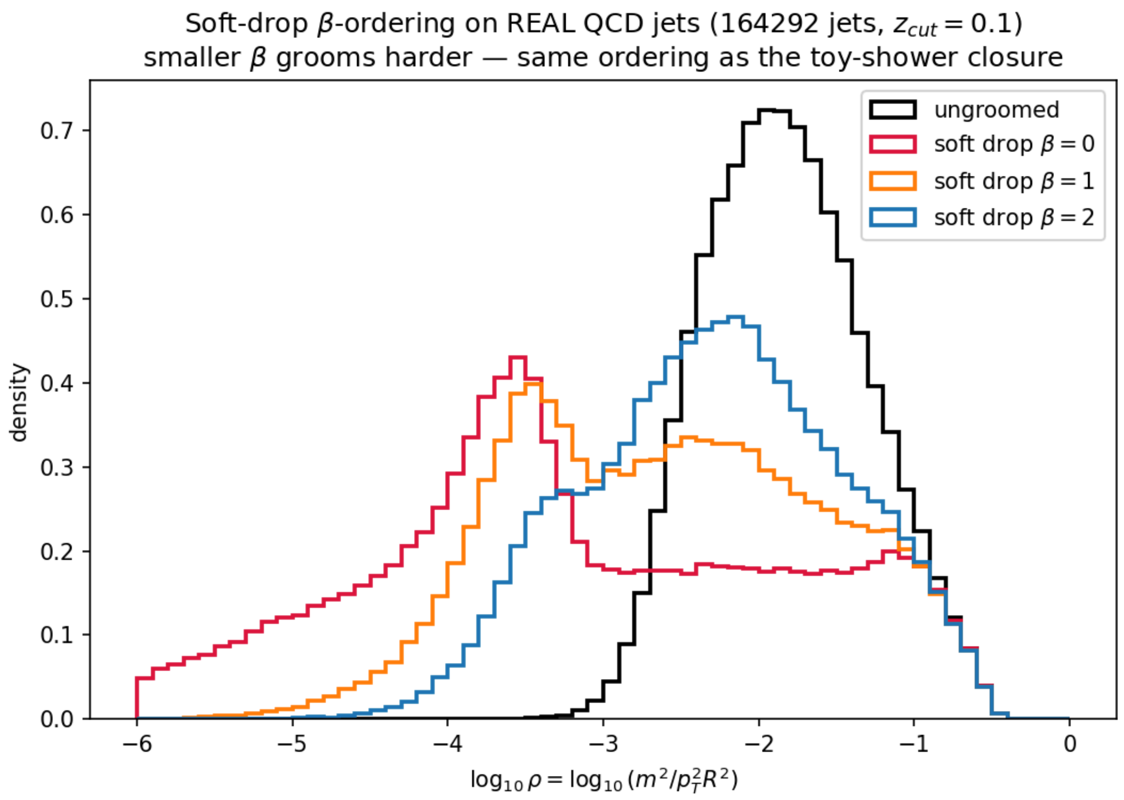
sample	p_T **ratio**	m_{SD} **Δ**	R_g **|Δ|<0.01**
QCD UL18	0.999999	−0.004 (95.7%<0.5)	**99.2%**
$t\bar{t}$ 2ℓ2ν	0.999998	−0.033 (94.1%<0.5)	95.9%
$t\bar{t}$ 4q '24	0.999556	−0.077 (69.3%<0.5)	87.0%

Soft-drop β-family on real QCD jets (F2)

The toy-shower closure, repeated on **164 292 real CMS jets**: re-groom the same C/A trees at $\beta = 0, 1, 2$ ($z_{\text{cut}} = 0.1$) and plot $\rho = m^2/(p_T^2 R^2)$.

- grooming pushes mass **down**; **smaller β grooms harder** — the exact ordering of Soft Drop [1402.2657] Figs. 3–4;
- $\beta = 0$ (mMDT) develops the characteristic flat low-ρ tail.

Input (C) — real constituents; grooming re-run 3× on the *same* merge histories
(a pure post-read: no re-clustering needed, which is the point of the history design).



Backup

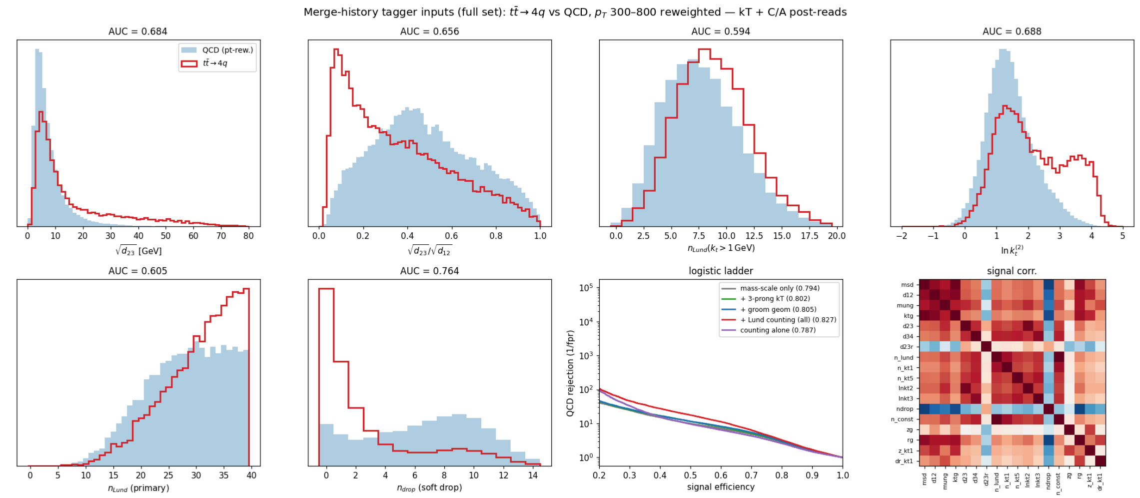
Outlook: history variables as tagger inputs

All of the above as an **18-variable input vector** — kT scales ($\sqrt{d_{12}}, \sqrt{d_{23}}, \sqrt{d_{34}}$),
C/A grooming ($m_{SD}, z_g, R_g, n_{drop}$), Lund summaries (emission counts, hardest $\ln k_t$'s) —
each a **post-read of histories we already have**.

$t\bar{t} \to 4q$ vs p_T-reweighted QCD, weighted logistic:

- mass-scale vars saturate at AUC **0.794** (0.8–0.97 correlated);
- the **declustering sequence** is the complement: full set **0.827**,
  ~2× QCD rejection at 30% eff;
- sleeper: n_{drop} alone 0.764 — decay jets pass soft drop in 0–1 declusterings;
- $\ln k_t^{(2)}$ resolves the **second** decay splitting.

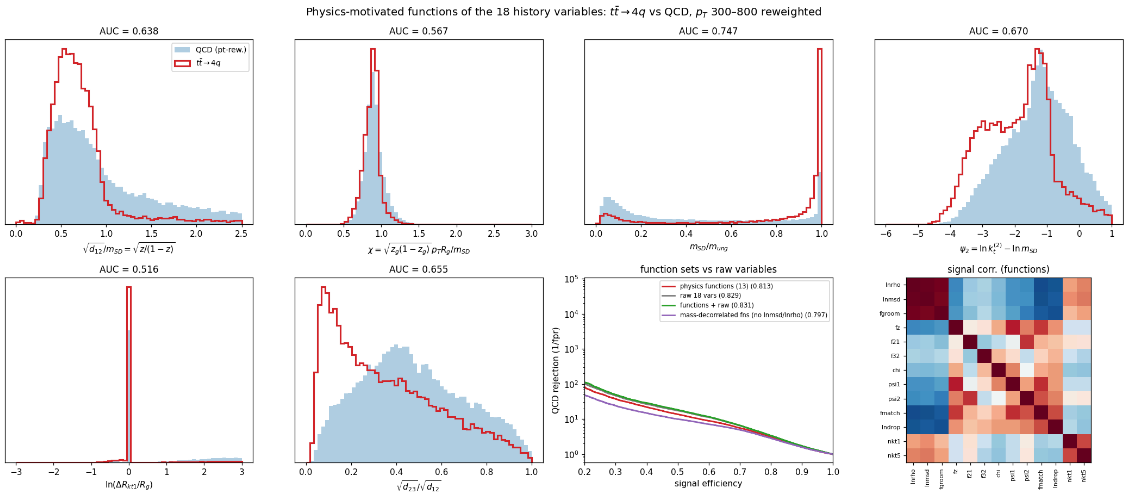
Exploratory (no gen-match, linear model, cross-era) — the point is the information is free once the history exists. Condor 9128460.



Outlook II: physics functions & the full history as tagger input

Compress to 13 dimensionless functions — lnρ, m_{SD}/m_{ung}, $\sqrt{d_{12}}/m_{SD} = \sqrt{z/(1-z)}$, prong ratios, closure $\chi = \sqrt{z_g(1-z_g)}\, p_T R_g / m_{SD}$ (<1 ⇒ massive prongs: top), $\psi_{1,2} = \ln k_t^{(1,2)} - \ln m_{SD}$, SD-split match, counting:

- functions-only **0.813**; with raw vars 0.831;
- **mass-decorrelated set: 0.797** with *no explicit mass input* — the sculpting-safe option.



Full history → **b-hive**: a merge history is a padded token sequence — exactly the cpf/npf/vtx input contract. C/A primary Lund list (= LundNet input) + kT scales
as two new groups; tree structure via ParT's pairwise channel. Config + ntuple branches +
~20 lines of `InputEmbed` — design note in the vault, not implemented.

Which tree does each variable come from?

Each jet is clustered **three times** (`extract_tagger_vars.py`), and every input is a post-read of one specific history:

anti-kT $R = 0.8$ — the physical jet

`algorithm="antikt"` → leading-jet 4-vector

- m_{ung} (ungroomed mass), n_{const}

kT **exclusive splitting scales**

`algorithm="kt"` → `splitting_scales()`

- $\sqrt{d_{12}}, \sqrt{d_{23}}, \sqrt{d_{34}}$ (value-sorted)
- ⇒ ratios d_{23}/d_{12}, f_{21}, f_{32}

C/A **big-R recluster → soft-drop + Lund**

`algorithm="cambridge"` → `groomed_jets` + `lund_coordinates`

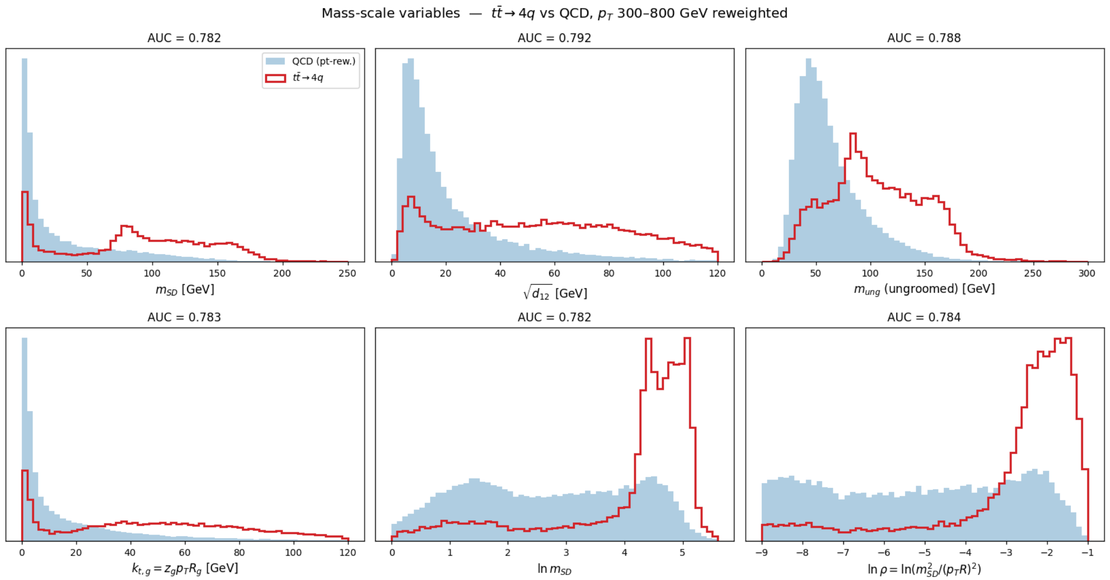
- groom: $m_{SD}, z_g, R_g, n_{drop}$, tagged
- Lund: $n_{Lund}, n(k_t > 1), n(k_t > 5), \ln k_t^{(1,2,3)}$, $z, \Delta R$ of hardest emission

mixed **functions spanning two trees**

- $f_{groom} = m_{SD}/m_{ung}$ (C/A ÷ anti-kT)
- $f_z = \sqrt{d_{12}}/m_{SD}$ (kT ÷ C/A)

Why two trees? kT is *value-sorted* — its exclusive scales $\sqrt{d_{12}} \geq \sqrt{d_{23}} \geq \ldots$ read off the hardest splittings directly (prong hierarchy). C/A is *angular-ordered* — its primary declustering **is** the Lund plane / soft-drop sequence (grooming, emission counts). anti-kT gives the jet itself. Every tag below (kT / C/A / anti-kT / mixed) marks the source.

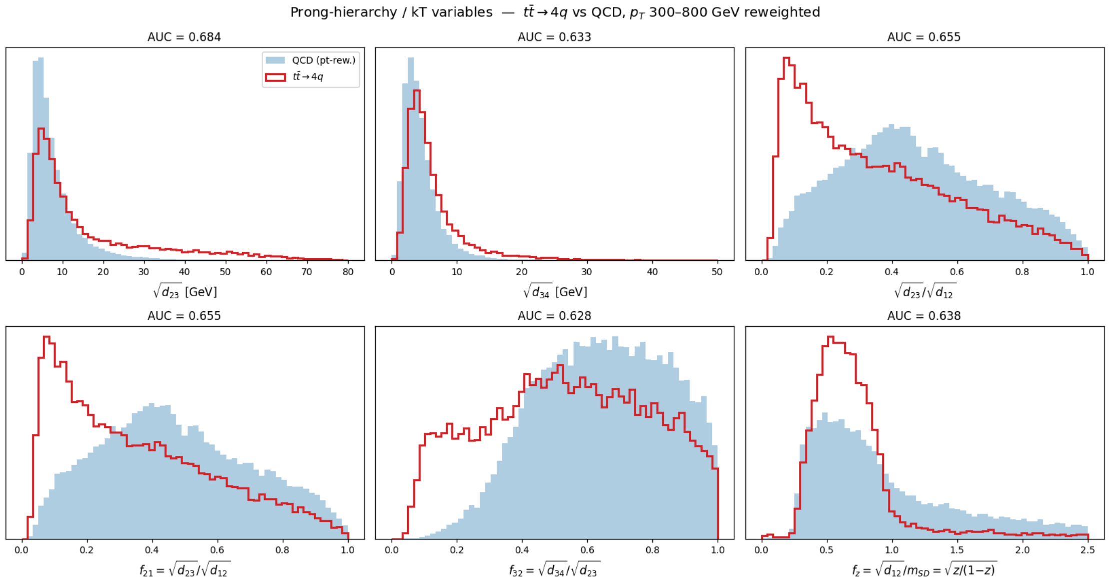
All inputs (1/5) — mass-scale variables



Filled = QCD (pt-reweighted), red = $t\bar{t} \to 4q$; per-panel weighted single-variable AUC. All at **0.78–0.79** and 0.8–0.97 correlated — they probe the *same* 2-prong decay mass, so add little to one another.

- C/A m_{SD} (0.782): soft-drop mass — clean $m_W \approx 80$ peak + $m_t \approx 160$ shoulder; QCD a steep low-mass continuum.
- kT $\sqrt{d_{12}}$ (0.792, best of group): $\approx \min(p_{T1}, p_{T2})\Delta R$ of the last merge — momentum-weighted mass scale of the hardest split.
- anti-kT m_{ung} (0.788): ungroomed mass — signal keeps its mass, QCD's is inflated by soft wide radiation.
- C/A $k_{t,g} = z_g p_T R_g$ (0.783): k_t of the groomed split, another mass proxy.
- C/A $\ln m_{SD}$ (0.782), $\ln \rho = \ln(m_{SD}^2/(p_T R)^2)$ (0.784): log forms. QCD is ~flat in $\ln \rho$; signal piles at the decay mass — cleanest of the group.

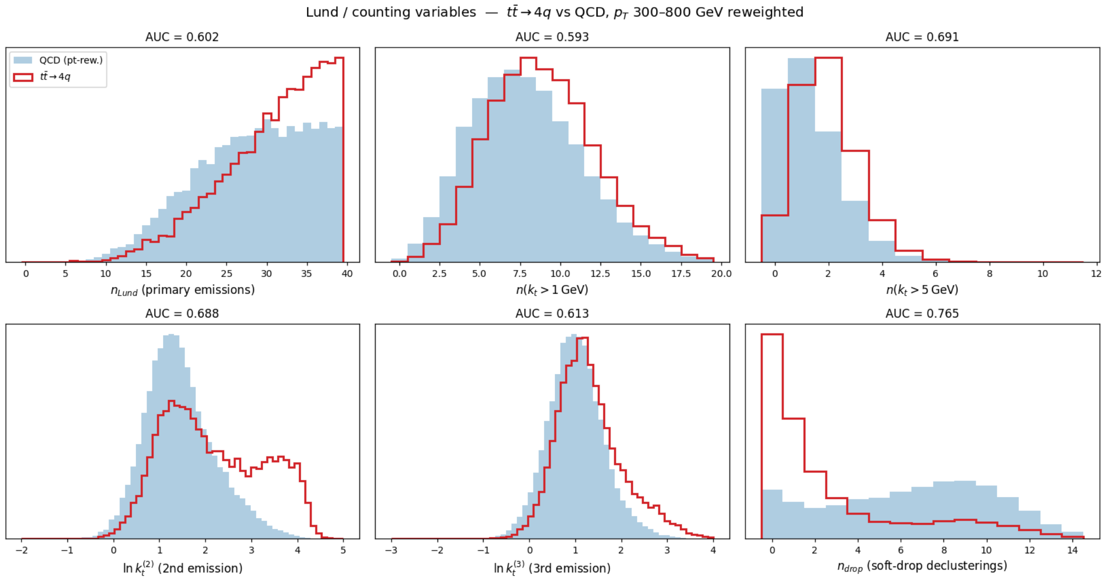
All inputs (2/5) — prong-hierarchy / kT variables



The kT splitting scales *beyond the first* and their **ratios** — do multiple hard prongs exist with a hierarchy (3-prong top, 2-prong W) vs QCD's single DGLAP ladder?

- kT $\sqrt{d_{23}}$ (0.684): 2nd kT scale — populated for 3-prong top (W→qq̄ inside), near zero for 2-prong W or 1-prong QCD.
- kT $\sqrt{d_{34}}$ (0.633): 3rd splitting — weakest, mostly extra radiation.
- kT d_{23}/d_{12} = f_{21} (0.655): the ratio removes overall scale — signal peaks near 0.1 (hierarchical decay scales), QCD broad.
- kT $f_{32} = \sqrt{d_{34}}/\sqrt{d_{23}}$ (0.628): next ratio in the hierarchy.
- kT÷C/A $f_z = \sqrt{d_{12}}/m_{SD} = \sqrt{z/(1-z)}$ (0.638): momentum sharing of the mass split, **mass-decorrelated** — decay shares evenly ($z \approx \frac{1}{2}$), QCD soft-biased.

All inputs (3/5) — Lund / counting variables

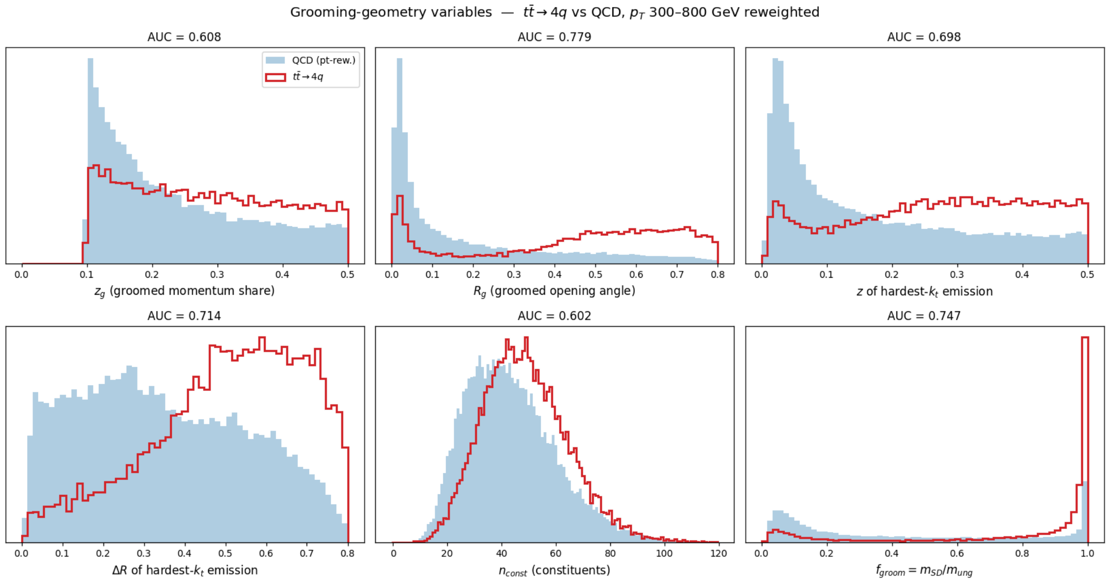


How many hard emissions, and how hard the sub-leading ones are — the **declustering sequence**, the information a single mass number cannot carry.

All from the **C/A** primary declustering sequence (the Lund plane + soft-drop walk).

- C/A n_{Lund} (0.602), $n(k_t > 1)$ (0.593): primary-emission counts. Signal has *fewer* (a color singlet radiates less) despite higher pileup — physical.
- C/A $n(k_t > 5)$ (0.691): counting only *hard* emissions sharpens it — top/W give 1–2, QCD more.
- C/A $\ln k_t^{(2)}$ (0.688): 2nd-hardest emission — signal bump at $\ln k_t \approx 3.5$–4.5 = the **second decay splitting** (top→W→qq̄).
- C/A $\ln k_t^{(3)}$ (0.613): 3rd emission — weaker.
- C/A n_{drop} (0.765, **best non-mass var**): soft-drop declustering count. Decays pass in **0–1** steps; QCD needs up to ~12.

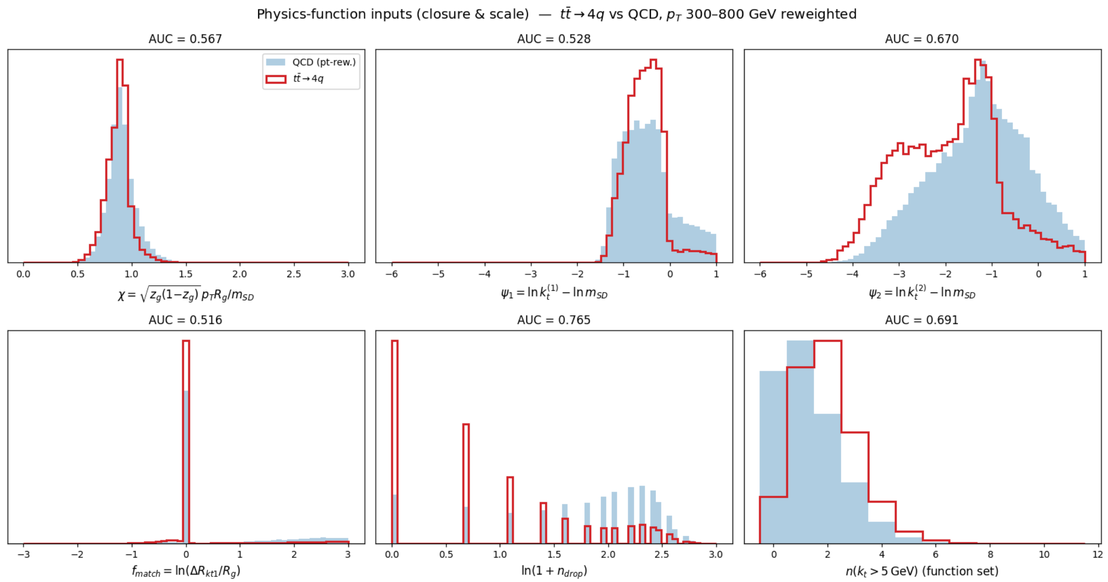
All inputs (4/5) — grooming-geometry variables



Geometry of the split soft drop keeps, plus constituent count and grooming survival.

- C/A z_g (0.608): groomed momentum share — near the $z_{cut} = 0.1$ edge, flatter than QCD's $1/z_g$; modest alone.
- C/A R_g (0.779): groomed opening angle, **strong**. Decay angle $R_g \approx m/(p_T\sqrt{z(1-z)})$ is fixed and wide; QCD collinear.
- C/A z (0.698), ΔR (0.714) **of the hardest-k_t emission**: for a decay this *is* the decay split, so both are decay-scale.
- anti-kT n_{const} (0.602): constituent multiplicity — quark/gluon-like, weak alone.
- C/A÷anti-kT $f_{groom} = m_{SD}/m_{ung}$ (0.747): **grooming survival** — decays keep mass (≈ 1), QCD loses it ($\ll 1$). Strong, mass-shape-decorrelated.

All inputs (5/5) — physics-function inputs



Dimensionless **closure / scale-comparison** functions, each designed to encode a specific decay hypothesis directly.

All AUCs weighted-logistic single-variable, same selection/reweighting as the study. Scripts `tagger_allvars.py` (plots), `tagger_study.py`, `tagger_functions.py`; Condor 9128460 extraction. See [[2026-07-18-tagger-inputs]].

All from the **C/A** groom + Lund reads (no kT tree enters this set).

- C/A $\chi = \sqrt{z_g(1-z_g)}\,p_T R_g/m_{SD}$ (0.567): 2-prong **closure** — $\chi\approx 1$ for massless prongs (W), $\chi < 1$ for **massive** prongs (top → Wb). Built-in top/W handle.
- C/A $\psi_1 = \ln k_t^{(1)} - \ln m_{SD}$ (0.528): hardest emission at the decay scale? Weak (≈ mass split for both).
- C/A $\psi_2 = \ln k_t^{(2)} - \ln m_{SD}$ (0.670): **2nd** emission at the decay scale? High for top (the W sub-decay) — strongest ψ/χ.
- C/A $f_{match} = \ln(\Delta R_{kt1}/R_g)$ (0.516): hardest-k_t emission ≡ SD split? ≈ 0 for a clean decay.
- C/A $\ln(1+n_{drop})$ (0.765): the n_{drop} handle, compressed.
- C/A $n(k_t > 5)$ (0.691): perturbative activity, in the function set.

Reproducibility & full-statistics numbers

All CMS plots regenerated **raw-to-raw on the C/A tree** at full statistics
(`make_cms_plots.py`, vectorized loader; run on **HTCondor** cluster 9087059, 60 257 jets):

observable	comparison	result
jet p_T (QCD)	our anti-kt vs `FatJet_pt×(1−rawFactor)`	median **1.000000**, σ 2.5×10^{-4}
soft-drop mass (QCD)	our C/A-tree vs m(raw sub_1+sub_2)	median **−0.004 GeV**, 95.7% <0.5 GeV
z_g (QCD)	our vs raw subjet z	$\lvert\Delta\rvert$ = **7×10^{-5}**
jet p_T (ttbar 2L2Nu)	12 561 leading jets, HTCondor 9098883	median **1.000002**, σ 2.2×10^{-4}
soft-drop mass (ttbar)	our C/A-tree vs m(raw sub_1+sub_2)	median **−0.041 GeV**, 94.2% <0.5 GeV
full-event (QCD)	all PFCands → anti-kt, ΔR-match to `FatJet_*`	7 701 jets, pt **1.0000**, ΔR med **0.0019**
R_g (3 samples)	our groomed `dR` vs ΔR(sub_1,sub_2), HTCondor 9099026	median Δ ≤ 2×10^{-4}; 99.2/95.9/87.0% <0.01
TTto4Q Run 3 2024	16 516 leading jets, raw-to-raw	pt 0.999556, m_{SD} −0.077 GeV — table-floor effect ↑ w/ pileup, rel. 0.14%

Residual = NanoAOD float storage throughout. Jets: anti-kt $R = 0.8$; grooming/Lund: big-R **C/A** reclustering (as FastJet).

Function reference (added)

```
exclusive_jets_from_history(hist_p1, hist_p2, hist_child, hist_d, mask,
                            n_jets=None, d_cut=None)          # F1
groom_from_history(hist_p1, hist_p2, hist_child, hist_d, mask, p4, R,
                   z_cut=0.1, beta=0.0, mu=None)              # F2 (soft-drop / mMDT / mass-drop)
lund_coordinates_from_history(hist_p1, hist_p2, hist_child, hist_d, mask, p4)  # F3 -> (B,J,S,6)

# helpers
_resolve_roots(...)      # pointer-jump roots of an arbitrary sub-forest
_jet_roots(...)          # per-jet root pseudojet id, beam-merge order
_pseudojet_p4(...)       # E-scheme 4-momentum of every pseudojet, keyed by id
_dense_number_roots(...) # compact per-particle jet index
```

API surface: `ClusterOutput.exclusive_jets`, `.groomed_jets`, `.mass_drop`, `.lund_coordinates`, plus a new `.mask` field to map slots→ids.

Reproducing everything

```
# on lxplus, env b_hive (torch 2.5.1)
cd /eos/home-c/cgupta/flashjet/plots/2026-07-13-substructure
micromamba run -n b_hive python make_plots.py        # F1/F2/F3 toy justification + parity
micromamba run -n b_hive python make_paper_plots.py  # anti-kt/SoftDrop/Lund paper figures
micromamba run -n b_hive python make_cms_plots.py    # real CMS UL18 QCD (needs qcd_jmenano_150x.root)
micromamba run -n b_hive python make_fullevent_plots.py  # full-event clustering vs stored AK8
micromamba run -n b_hive python make_ttbar_plots.py  # ttbar (needs ttbar_jmenano_150x.root)
micromamba run -n b_hive python make_compare_plots.py # QCD vs 2L2Nu vs TTto4Q comparisons
micromamba run -n b_hive python outliers.py          # m_SD outlier anatomy
# long jobs run on HTCondor: submit dir ~/flashjet_condor (AFS; /eos paths are
# rejected in submit files), executables cd to EOS + `micromamba run -n b_hive`

# tests
cd /eos/home-c/cgupta/flashjet/FlastJetDemo
micromamba run -n b_hive python -m pytest -q          # 85 passed, 13 skipped
```

Fixed seed `20260713` throughout. Papers catalogued in `References/Flashjet/papers.md`.